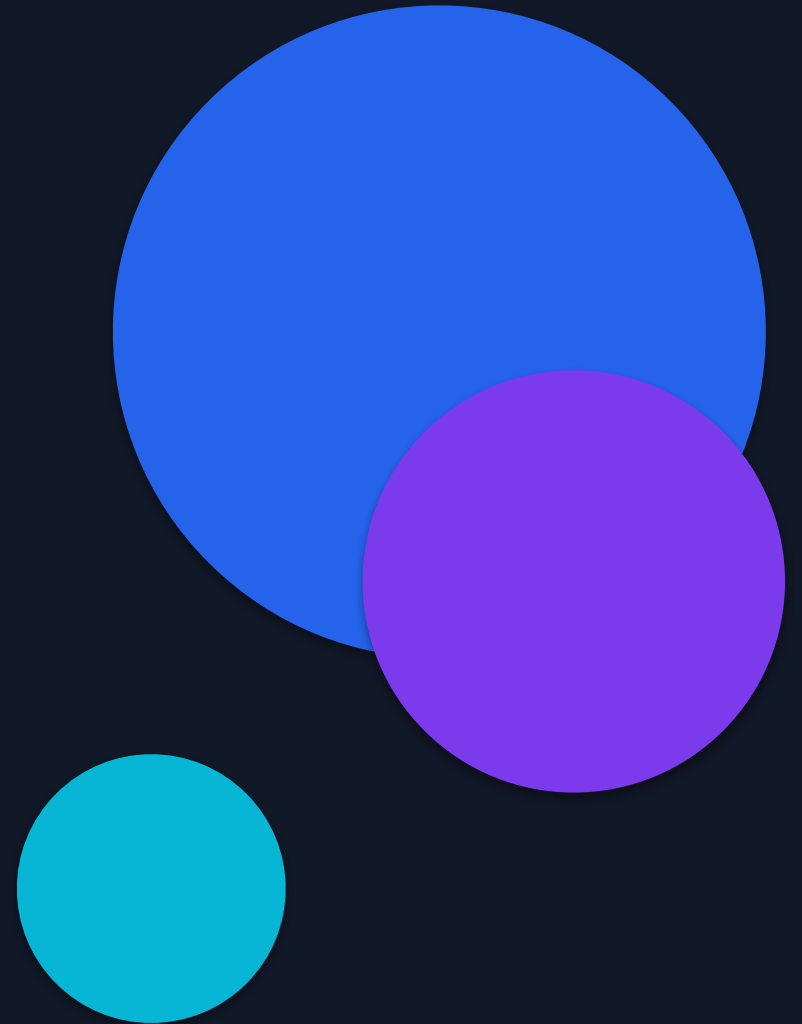


**Prof. Bishal Sarkar**

# **Generative AI & Agentic AI**



Lecture 01 • AI, ML & Generative AI: The Big Picture

# AI, ML & Generative AI: The Big Picture

Today's goal: build the mental map before learning the machinery.

## AI Artificial Intelligence

The broad field: making machines perform tasks associated with intelligence.

## ML Machine Learning

Systems learn patterns from data rather than relying only on hand-written rules.

## Gen AI Generative AI

Models that can create new content such as text, images, audio, video and code.

**Question for today: how do these ideas fit together?**

# How many times did you interact with AI today?

Think beyond ChatGPT.

- Google Maps / navigation
- YouTube / Netflix recommendations
- Face unlock / photo organization
- Spam & fraud detection
- Search, translation, autocorrect
- Generative tools for text, images or code

## Think about it

If AI is already embedded in everyday products, the real question is not “Will AI arrive?”

It is: “Where is AI already working — and what kind of AI is it?”

**The question we will answer today**

**AI • ML • Deep Learning • Generative AI • Agentic AI**

**Different technologies — or different layers of one bigger idea?**

# What is Artificial Intelligence?

A useful classroom definition.

**AI is the field of building machines that can perform tasks requiring capabilities we associate with intelligence.**

## 1 Perceive

See, hear, sense

## 2 Reason

Infer, compare, plan

## 3 Learn

Improve from experience

## 4 Act

Recommend, decide, execute

# AI is not one algorithm

It is a large family of approaches.

## Rule-based AI

Humans specify rules.

Example: if temperature > threshold → trigger alert.

## Machine Learning

The system learns patterns from examples.

Example: spam detection.

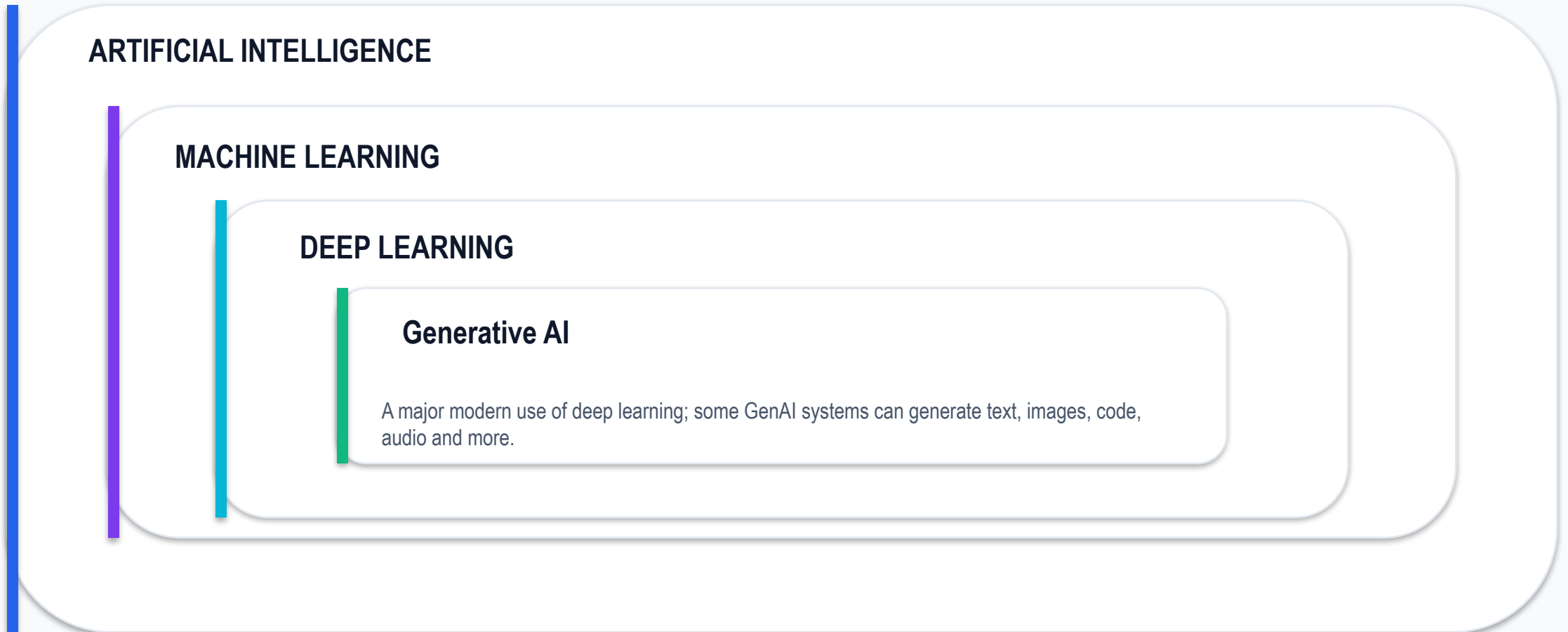
## Modern Generative AI

The model learns rich representations and can generate new content.

**The methods differ — the goal is intelligent behavior.**

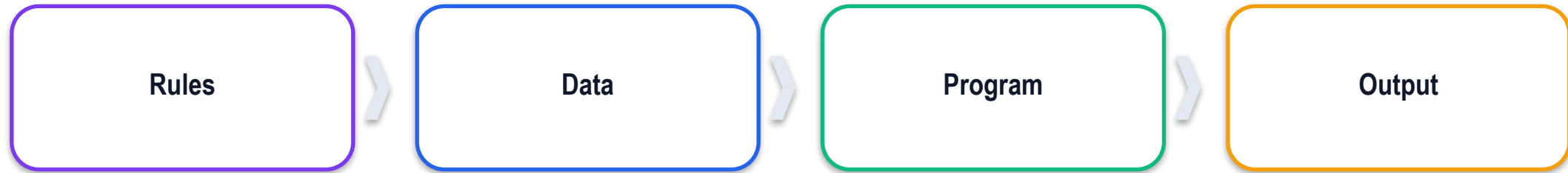
# The mental map

A practical way to organize the terminology.



# Before Machine Learning

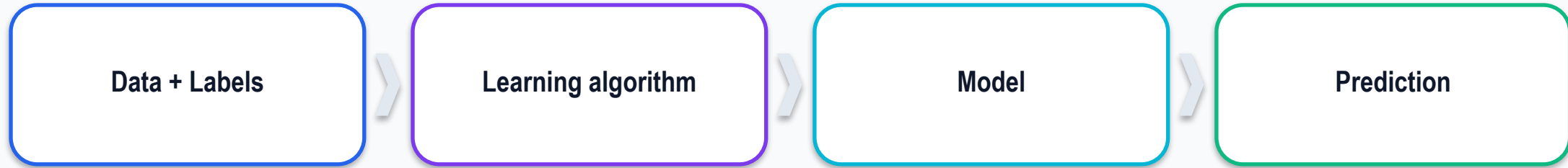
Traditional programming works well when the rules are known.



**Challenge: what if the rules are too numerous, fuzzy, or constantly changing?**



# Machine Learning: learn from examples



## Core idea

Instead of programming every rule, we give the system examples and let it discover useful patterns.

# Example: teach a computer to detect spam

Can we write one perfect rule for every spam email?

| Email example                       | Label    |
|-------------------------------------|----------|
| “Congratulations! You won ₹10 lakh” | SPAM     |
| “Meeting at 3 PM”                   | NOT SPAM |
| “Claim your reward now”             | SPAM     |
| “Assignment submission tomorrow”    | NOT SPAM |

**The model learns patterns associated with “spam” — not a single hard-coded rule.**

# Three common ways machines learn

You will revisit these in detail later.

## 1 Supervised Learning

We have examples with known answers.

Examples:

- spam detection
- house-price prediction
- disease classification

## 2 Unsupervised Learning

We have data, but no target label.

Examples:

- customer segmentation
- anomaly discovery
- dimensionality reduction

## 3 Reinforcement Learning

An agent learns through actions, rewards and feedback.

Examples:

- games
- robotics
- sequential decision-making

# Prediction is not the same as intelligence

A model can be excellent at one task without being generally intelligent.

## Predictive ML

Input → model → prediction

“Will this transaction be fraudulent?”

“Will this customer churn?”

“What will demand be next week?”

## Decision / Action

Goal → options → decision → action

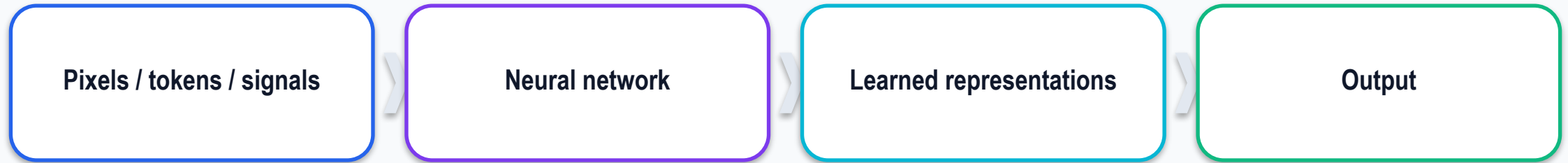
“What should we do next?”

This becomes especially important when we discuss agents.

**Prediction estimates. Decision-making chooses.**

# Deep Learning: neural networks at scale

Deep learning uses multi-layer neural networks to learn complex representations.



## Vision

Image → features → object

## Aa Language

Tokens → representations → meaning



## Speech

Audio → patterns → words

# Why did Deep Learning become so powerful?

Three forces reinforced each other.

## More Data

Digital products generate huge amounts of text, images, video, clicks and sensor data.

## More Compute

GPUs and specialized accelerators made large-scale training practical.

## Better Architectures

Modern neural-network architectures improved representation learning and scaling.

**Data × Compute × Algorithms → a new era of AI**

# Generative AI: from predicting to creating

**Generative AI produces new content that resembles patterns learned during training.**

## **Text**

Emails, stories, explanations

## **Images**

Designs, illustrations

## **Code**

Functions, tests, documentation

## **Audio / Video**

Speech, music, video

# Traditional AI vs Generative AI

## → Traditional / Predictive AI

Input → model → score / class / prediction

Examples:

- Fraud / not fraud
- Spam / not spam
- Demand forecast
- Recommendation

## ◆ Generative AI

Prompt / context → model → new content

Examples:

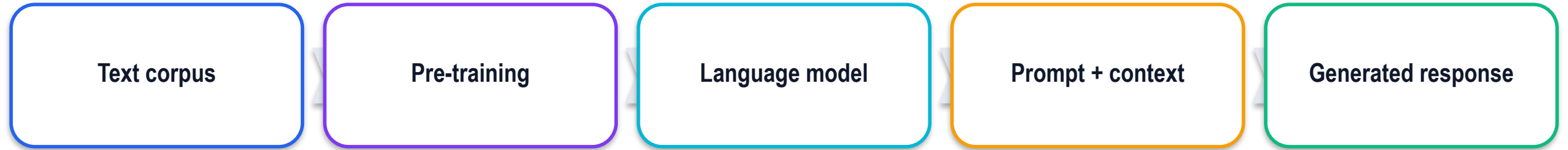
- Write an email
- Generate an image
- Summarize a document
- Create code

**Predict • Classify • Recommend ↔ Generate • Transform • Create**



# Large Language Models: the engine behind modern text GenAI

A simplified view — not the full training story.



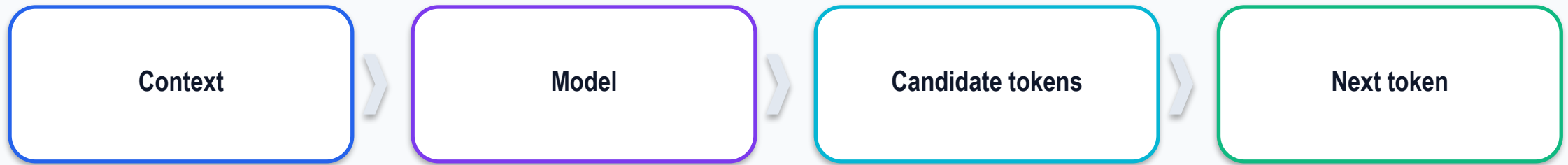
## LLM Important intuition

The model learns statistical structure in language and uses that learned representation to generate a continuation conditioned on the context.

# One useful mental model: next-token prediction

This is an intuition, not the whole story of an LLM.

**“The capital of France is ...”**



**Paris • likely continuation**

# Agentic AI: when the model starts pursuing a goal

**Generative AI creates responses.**  
**Agentic AI can use models + tools + memory + feedback to accomplish goals.**

## Goal

Understand what the user wants

## Plan

Break the goal into steps

## Act

Call tools / APIs / systems

## Check

Observe results and adapt

# Example: “Book me the cheapest flight to Delhi tomorrow morning.”



## Agent      What makes this agentic?

The system is not only generating text. It is taking actions, observing results and moving toward a user-defined outcome.

# Generative AI vs Agentic AI

## Generative AI

Primary capability: generate / transform content

Typical interaction:

User → prompt → model → response

Examples:

- Draft an email
- Explain a concept
- Generate code

## Agentic AI

Primary capability: pursue a goal through actions

Typical interaction:

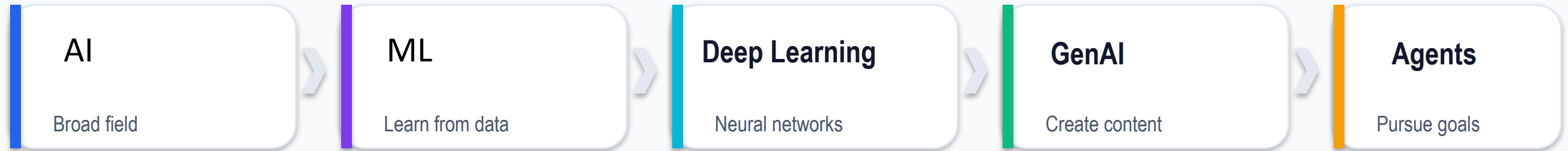
Goal → plan → tools → observe → adapt

Examples:

- Research and summarize
- Schedule a workflow
- Execute multi-step tasks

**An agent can use GenAI — but GenAI does not automatically make a system an agent.**

# The big picture — one mental model



## ★ Nuance to remember

AI → ML → Deep Learning is a useful hierarchy.

Generative AI and Agentic AI are capabilities / system patterns that can build on modern ML and deep learning — they are not simply extra boxes in one strict nesting hierarchy.

# Where modern AI fits in the industry



## Banking

Fraud detection • credit risk • support



## Healthcare

Imaging • documentation • decision support



## E-commerce

Recommendations • search • personalization



## Manufacturing

Predictive maintenance • quality inspection



## Education

Tutoring • content generation • assessment

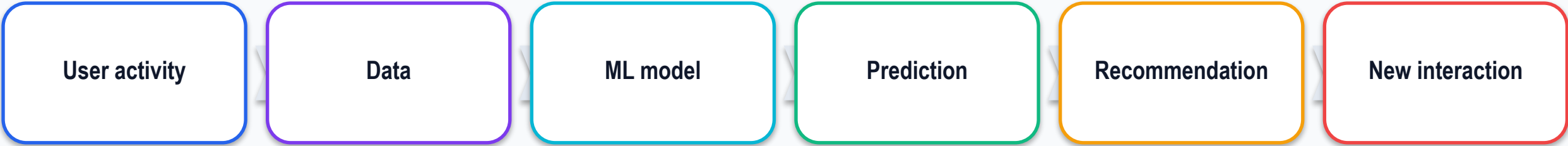


## Software

Code generation • testing • developer copilots

# Real-world workflow: product recommendation

Where do AI and ML appear in a familiar product?



**ML**

Learns patterns from historical interactions.

**Prediction**

Estimates what the user may prefer.

**Feedback loop**

The next interaction becomes new data.



# Mini Challenge

Work in pairs / small groups. You have 3 minutes.

**Classify each scenario by its dominant AI capability.**

## AI / ML / DL

Which is the best label?

## Generative AI

Is new content being created?

## Agentic AI

Is the system pursuing a goal through actions?

## Mini Challenge: classify these 5 problems

**A**  
Gmail detects spam automatically.

**B**  
Netflix recommends a movie you may like.

**C**  
ChatGPT writes a professional email from your notes.

**D**  
A self-driving car detects a pedestrian.

**E**  
An AI system searches hotels, compares options and completes a booking.

# Challenge: discuss the reasoning, not just the label

A  
**Machine Learning**

Learned pattern → spam / not spam

B  
**Machine Learning**

Prediction / ranking of likely preferences

C  
**Generative AI**

Creates new text from instructions and context

D  
**Deep Learning**

Vision model detects and localizes objects

E  
**Agentic AI**

Plans, uses tools, observes results and acts

# Five ideas to take away

## 1 AI is the broad field

Machines performing tasks associated with intelligence.

## 2 ML learns patterns

Data + learning algorithm → model.

## 3 Deep Learning scales learning

Neural networks learn rich representations.

## 4 GenAI creates

Text, images, code, audio, video and more.

## 5 Agentic AI acts toward goals

Plan → tools → observe → adapt.

## One final question

**If AI can predict,  
generate and act...**

**what should humans still be responsible for?**

Next: we move from the big picture to how ML actually works.